LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING

GitHub to Disable npm Install Scripts by Default to Stop Supply Chain Attacks

Vulnerability Report

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Author	Lost Boys Cyber
Report Type	Vulnerability Report

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1. Executive Summary

GitHub has announced a set of npm v12 security-default changes intended to reduce software supply-chain compromise paths that rely on automatic code execution during `npm install`. The most consequential change is that dependency lifecycle scripts (`preinstall`, `install`, and `postinstall`) will no longer run by default unless the project explicitly approves them. Git dependencies and remote URL dependencies will also require explicit opt-in through `--allow-git` and `--allow-remote` controls.

This is not a CVE-specific vulnerability disclosure. It is a platform hardening change affecting one of the largest recurring execution surfaces in JavaScript development: install-time code run from direct and transitive dependencies on developer endpoints and CI/CD runners. For defenders, the change is strategically positive because it weakens a common initial execution path used by malicious packages, compromised maintainers, dependency confusion, and poisoned transitive dependencies.

The operational risk is two-sided. Organizations that prepare early can reduce exposure to malicious install scripts while preserving trusted native-build workflows. Organizations that do not prepare may experience build instability when npm v12 lands, or may create overly broad allowlists that preserve the same risk the change is meant to reduce. Lost Boys Cyber assesses this item as medium severity and high confidence for organizations that build, package, or deploy Node.js software, including manufacturing environments with embedded web tooling, internal portals, OT-adjacent dashboards, vendor build chains, or CI/CD pipelines that consume npm dependencies.

Field	Assessment
Report ID	LBC-VTI-2026-079
Report Type	Vulnerability / Supply-Chain Threat Intelligence
Primary Change	npm v12 disables dependency install scripts by default unless approved
Additional Changes	Git dependencies and remote URL dependencies require explicit allow flags
Exposure Window	Preparation available in npm 11.16.0+; npm v12 estimated July 2026
Severity	Medium overall; High for CI/CD environments that install unpinned or weakly governed dependencies
Confidence	High for announced npm behavior; Medium for organization-specific impact until dependency audits are complete

2. Intelligence Requirement

This report addresses the following intelligence requirement: What does GitHub's npm v12 install-time hardening mean for defenders responsible for developer workstations, CI/CD runners, and manufacturing-sector software delivery environments?

Requirement	Analytic Answer
What changed?	npm v12 will make install-time dependency scripts opt-in instead of trusted-by-default.
Why does it matter?	A single compromised direct or transitive dependency can currently execute arbitrary code during `npm install` on developer machines or build runners.
Who is exposed?	Teams using npm in local development, CI/CD, container builds, release automation, firmware tooling, internal applications, and

	vendor-supplied JavaScript projects.
What should defenders do now?	Upgrade to npm 11.16.0+, review warnings, run `npm approve-scripts --allow-scripts-pending`, approve only trusted packages, commit the resulting allowlist, and hunt for suspicious install-time execution.
What is not known?	Organization-specific breakage depends on local dependency trees, native module usage, Git/remote URL dependencies, and existing `ignore-scripts` or package-manager policy settings.

Analyst assumptions: this report treats the GitHub changelog as the authoritative source for the npm v12 behavior and The Hacker News as secondary reporting that summarized the security rationale. No public evidence in the reviewed sources indicates active exploitation of a newly disclosed npm CLI vulnerability; the defensive relevance comes from reducing abuse of existing npm lifecycle behavior.

3. Source Review and Confidence

Source	Contribution	Reliability	Confidence
GitHub Changelog, 2026-06-09	Primary statement of npm v12 breaking changes, timeline, allowlist workflow, and affected install behaviors.	High	High
GitHub Community discussion	Clarifies operational behavior, including that npm v12 default skips unapproved scripts with warnings while stricter enforcement can hard-fail installs.	Medium-High	Medium-High
The Hacker News, 2026-06-11	Secondary security-media summary tying the change to supply-chain attack prevention and lifecycle-hook abuse.	Medium	Medium
Existing defender telemetry patterns	Informs detection and hunting recommendations for npm/node process trees, CI runners, and egress during package install.	Medium	Medium

Key source claims assessed as high confidence:

- npm v12 is expected in July 2026.
- npm 11.16.0+ already exposes warnings so maintainers can prepare before npm v12.
- `allowScripts` defaults to off in npm v12 for dependency lifecycle scripts unless explicitly approved.
- `--allow-git` and `--allow-remote` default to `none`, requiring explicit opt-in for Git and remote tarball dependency resolution.
- Native `node-gyp` builds and `prepare` scripts from Git, file, and link dependencies can be affected because npm may otherwise run implicit build or prepare steps.

Confidence caveats:

- No single CVE, exploit proof-of-concept, or named adversary campaign is attached to this hardening announcement.
- The report does not assert compromise in any specific client environment.
- Detection logic below is behavioral and should be tuned against approved build systems, package-manager baselines, and expected native-module compilation activity.

4. Key Findings

ID	Finding	Defensive Implication	Priority
F1	npm dependency lifecycle scripts are being moved from automatic execution to explicit approval.	This reduces a high-value execution path for malicious or compromised npm packages, especially transitive dependencies.	High
F2	CI/CD runners and developer workstations are the primary control points.	Endpoint and build telemetry should focus on `npm`, `node`, shell, compiler, network, and credential-access process relationships during installs.	High
F3	Git and remote URL dependencies require explicit review.	Repositories using Git dependencies, remote tarballs, or private package distribution shortcuts may break unless owners approve intentional usage.	Medium-High
F4	Native module workflows may be affected.	Packages that rely on `node-gyp` or postinstall downloads/builds require allowlist decisions and compensating controls.	Medium
F5	A broad allowlist can reintroduce the original risk.	Security teams should require package-level justification, source review, and owner signoff before approving install scripts.	High
F6	Existing `ignore-scripts=true` settings may override allowlist behavior.	Teams should inventory package-manager configuration so they understand whether scripts are blocked, skipped, or explicitly allowed.	Medium

Defensive interpretation: the change should be treated as a supply-chain control migration, not merely a developer-experience issue. Security teams should use the migration window to baseline which packages truly require install-time execution, remove unnecessary Git/remote URL dependencies, and alert on package installs that spawn shells, download tools, credential utilities, or unexpected outbound connections.

5. Threat Context

npm install-time execution has historically been attractive to adversaries because it combines trust inheritance with operational reach. A developer or CI runner executes a normal package-manager command, npm traverses the dependency tree, and lifecycle hooks can execute code before an application is ever run. This can give a malicious package access to source code, environment variables, build secrets, package-registry tokens, SSH keys, cloud credentials, and internal network paths.

Common abuse patterns relevant to this change include:

Abuse Pattern	Description	Defensive Relevance
Malicious dependency lifecycle hook	A package executes code from `preinstall`,	Directly addressed by npm v12

	`install`, or `postinstall`.	`allowScripts` default-off behavior.
Compromised transitive dependency	A package deep in the dependency tree executes code when installed by a trusted top-level project.	Reduces blast radius by requiring explicit approval instead of implicit trust.
Git dependency execution path	A dependency resolved from Git can carry configuration or prepare behavior that executes during install.	npm v12 requires `--allow-git`; defenders should minimize Git dependencies in production builds.
Remote tarball dependency	A dependency resolved from an HTTPS tarball bypasses standard registry review and provenance controls.	npm v12 requires `--allow-remote`; defenders should review remote URL usage.
Native module postinstall build/download	Packages compile native modules or fetch binaries during install.	Some legitimate packages need approval, but the pattern overlaps with malware staging and egress.
CI secret theft	Install-time code reads tokens and environment variables from build runners.	Detection should prioritize runner process trees and unusual outbound traffic during installs.

MITRE ATT&CK mapping for the underlying abuse path:

Technique	Name	Rationale
T1195.001	Supply Chain Compromise: Compromise Software Dependencies and Development Tools	Malicious npm packages or compromised dependencies can enter the build chain through package installation.
T1059.004	Command and Scripting Interpreter: Unix Shell	Install scripts frequently spawn `sh`, `bash`, or platform shells to run payload logic.
T1059.007	Command and Scripting Interpreter: JavaScript	npm and Node.js provide JavaScript execution as the package-manager runtime.
T1105	Ingress Tool Transfer	Malicious install scripts may download second-stage code, binaries, or configuration.
T1552.001	Unsecured Credentials: Credentials In Files	Install-time code on developer or CI systems may read `.npmrc`, SSH keys, cloud credentials, or token files.
T1041	Exfiltration Over C2 Channel	Compromised build jobs may exfiltrate tokens or source data over outbound HTTPS/DNS.

Manufacturing-sector relevance is strongest where production engineering, plant-support applications, internal dashboards, vendor portals, firmware build systems, or OT-adjacent integration layers use JavaScript build tooling. Even when npm is not deployed on industrial control assets, compromised developer or CI/CD systems can become a path into source repositories, signing material, deployment automation, or vendor-delivered software artifacts.

Detection coverage priorities

Analytic	Telemetry Source	Goal	Notes
npm install spawns shell and network tooling	Endpoint process telemetry, EDR, CI runner logs	Identify install-time execution that downloads or stages	Tune approved packages that legitimately build native

		payloads.	modules.
npm/node reads credential files during install	File telemetry, EDR	Detect credential harvesting from `.npmrc`, SSH, cloud, and CI token paths.	Prioritize developer endpoints and self-hosted runners.
npm install creates persistence or modifies shell profile	Endpoint process/file telemetry	Detect malicious postinstall persistence attempts.	Rare in legitimate builds; high signal when observed.
Git or remote URL dependencies in manifests	Source control scanning, dependency manifests	Identify projects likely to require npm v12 opt-in flags.	Use as migration inventory and policy gate.
New warnings from npm 11.16.0+	CI logs, build artifacts	Find packages that will be skipped under npm v12.	Convert warnings into package-owner review tasks.

Sigma: suspicious npm install-time child process

```
title: Suspicious Child Process Spawned During NPM Install
id: 5d7f4a30-2ff0-49dd-93d5-npm-install-child-process
status: experimental
description: Detects npm or node process trees spawning shells, download utilities, or credential-
access tooling during dependency installation.
author: Lost Boys Cyber
date: 2026-06-16
references:
  - https://github.blog/changelog/2026-06-09-upcoming-breaking-changes-for-npm-v12/
  - https://thehackernews.com/2026/06/github-to-disable-npm-install-scripts.html
tags:
  - attack.t1195.001
  - attack.t1059
  - attack.t1105
logsource:
  category: process_creation
detection:
  parent_npm:
    ParentImage|endswith:
      - '/npm'
      - '/npm-cli.js'
      - '\\npm.cmd'
      - '\\node.exe'
  suspicious_child:
    Image|endswith:
      - '/sh'
      - '/bash'
      - '/zsh'
      - '/curl'
      - '/wget'
      - '/python'
      - '/python3'
      - '/powershell.exe'
      - '/pwsh'
      - '\\cmd.exe'
      - '\\powershell.exe'
      - '\\curl.exe'
      - '\\wget.exe'
  install_context:
    CommandLine|contains:
      - 'npm install'
      - 'npm ci'
      - 'preinstall'
      - 'postinstall'
      - 'node-gyp'
  condition: parent_npm and suspicious_child and install_context
falsepositives:
```

```
  - Legitimate native module compilation or approved package postinstall scripts.
  - Internal build scripts that intentionally fetch platform-specific binaries.
level: medium
```

Microsoft Defender XDR hunt: npm install child processes and outbound staging

```
let Lookback = 30d;
let NpmParents = DeviceProcessEvents
| where Timestamp > ago(Lookback)
| where FileName in~ ('npm', 'npm.cmd', 'npm-cli.js', 'node', 'node.exe')
   or ProcessCommandLine has_any ('npm install', 'npm ci', 'approve-scripts', 'allow-scripts');
let SuspiciousChildren = DeviceProcessEvents
| where Timestamp > ago(Lookback)
| where InitiatingProcessFileName in~ ('npm', 'npm.cmd', 'npm-cli.js', 'node', 'node.exe')
| where FileName in~
('sh','bash','zsh','curl','wget','python','python3','powershell.exe','pwsh','cmd.exe','certutil.exe')
   or ProcessCommandLine has_any ('preinstall','postinstall','node-gyp','/tmp/','Invoke-
WebRequest','iwr','curl ','wget ');
SuspiciousChildren
| project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName,
InitiatingProcessCommandLine, ReportId, DeviceId
| join kind=leftouter (
    DeviceNetworkEvents
    | where Timestamp > ago(Lookback)
    | where InitiatingProcessFileName in~
('npm','npm.cmd','node','node.exe','curl','wget','powershell.exe','pwsh','python','python3')
    | project NetTime=Timestamp, DeviceId, RemoteUrl, RemoteIP, RemotePort,
InitiatingProcessCommandLine
) on DeviceId
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), RemoteUrls=make_set(RemoteUrl, 20),
RemoteIPs=make_set(RemoteIP, 20), ExampleCommand=any(ProcessCommandLine) by DeviceName, AccountName,
FileName, InitiatingProcessFileName
| order by LastSeen desc
```

Microsoft Defender XDR hunt: package manifest dependencies requiring npm v12 review

```
DeviceFileEvents
| where Timestamp > ago(30d)
| where FileName =~ 'package.json'
| where ActionType in ('FileCreated', 'FileModified')
| join kind=inner (
    DeviceTvmSoftwareEvidenceBeta
    | project DeviceId, SoftwareName, SoftwareVersion
) on DeviceId
| project Timestamp, DeviceName, FolderPath, InitiatingProcessFileName, InitiatingProcessCommandLine,
SoftwareName, SoftwareVersion
| where InitiatingProcessCommandLine has_any ('git+', 'github:', 'http://', 'https://', 'file:',
'link:')
```

If Defender file-content telemetry is unavailable, run the equivalent source-control query in GitHub code search or CI preflight:

```
rg -n '"(dependencies|devDependencies|optionalDependencies)"|git\+|github:|https?://|file:|link:' --
glob 'package.json' .
```

Splunk hunt: npm install script process chain

```
index=edr sourcetype=process earliest=-30d
(parent_process_name IN (npm,npm.cmd,node,node.exe,npm-cli.js) OR process_command_line="*npm install*"
OR process_command_line="*npm ci*")
(process_name IN (sh,bash,zsh,curl,wget,python,python3,powershell.exe,pwsh,cmd.exe,certutil.exe) OR
process_command_line="*postinstall*" OR process_command_line="*preinstall*" OR
process_command_line="*node-gyp*")
| stats earliest(_time) as first_seen latest(_time) as last_seen values(process_command_line) as
commands values(dest) as hosts by host user parent_process_name process_name
| convert ctime(first_seen) ctime(last_seen)
| sort - last_seen
```

Hunt triage questions:

- Was the script spawned from a dependency lifecycle hook or an intentional repository script?
- Was the package already approved through `npm approve-scripts`, and is the approval justified by an owner?
- Did the process contact new infrastructure, paste services, object storage, raw GitHub URLs, or unapproved package mirrors?
- Did it read `.npmrc`, `.ssh`, cloud credential files, CI environment variables, or repository secrets?
- Did the activity happen on a developer endpoint, shared build runner, release runner, or production deployment host?

7. Recommendations

Priority	Owner	Recommendation	Target State
Critical	AppSec / Platform Engineering	Upgrade representative CI and developer test environments to npm 11.16.0+ and capture install warnings.	Known list of packages that require script approval before npm v12.
Critical	Repository Owners	Run `npm approve-scripts --allow-scripts-pending` in each actively maintained npm project and approve only packages with a documented business need.	Package-level allowlist committed to `package.json` with owner justification.
High	CI/CD Engineering	Inventory Git and remote URL dependencies and remove or replace them with registry-pinned packages where possible.	No unreviewed Git or remote tarball dependencies in release pipelines.
High	Security Engineering	Deploy behavioral detections for npm/node spawning shells, download tools, credential access, or unexpected network egress during installs.	Alerts routed to SOC/AppSec with CI runner context and repository metadata.
High	Governance / SDLC	Add npm v12 migration checks to pull-request templates, dependency review, and build policy.	New dependencies cannot introduce install scripts without explicit approval.
Medium	Developer Experience	Publish internal guidance explaining `allowScripts`, `approve-scripts`, `deny-scripts`, `ignore-scripts`, `--allow-git`, and `--allow-remote`.	Developers understand breakage patterns and do not bypass controls globally.
Medium	Incident Response	Update supply-chain playbooks to include npm install-time process trees, package-lock review, token rotation, and CI runner isolation steps.	Faster triage when suspicious package-install execution is detected.

Immediate preparation workflow:

```
npm --version
npm install -g npm@^11.16.0
npm install
npm approve-scripts --allow-scripts-pending
```

```
npm approve-scripts <trusted-package-name>
npm deny-scripts <untrusted-or-unneeded-package-name>
git diff -- package.json package-lock.json
```

Operational guardrails:

- Do not approve all packages globally to silence warnings.
- Require package-owner signoff for every install script that remains allowed.
- Treat native module packages as review candidates, not automatic exceptions.
- Prefer registry packages with provenance and lockfile integrity over Git or remote tarball dependencies.
- Preserve CI logs showing npm v12 warnings or skipped scripts for audit and follow-up.
- For self-hosted runners, isolate secrets so package-install jobs cannot read release credentials unless required.

Indicators of Compromise

No source-validated malicious atomic indicators were disclosed for this work item. Lost Boys Cyber therefore treats this section as an IOC disposition record so downstream publication, OpenCTI hygiene, and detection-engineering workflows do not create placeholder blocklist entries.

Indicator Type	Value	Disposition
Package-install execution behavior	No package names, domains, IPs, hashes, or URLs disclosed as malicious IOCs	Hunt for suspicious npm lifecycle script process trees and unapproved dependency behavior.
Workflow context	threat-intel / threat-intel-intake	Use the report's detection and hunting logic for monitoring; do not promote non-malicious metadata into IOC-only controls.

8. References

- [1] GitHub Changelog: Upcoming breaking changes for npm v12, 2026-06-09. https://github.blog/changelog/2026-06-09-upcoming-breaking-changes-for-npm-v12/
- [2] GitHub Community Discussion: Preparing for npm v12 install scripts and non-registry dependency changes. https://github.com/orgs/community/discussions/198547
- [3] The Hacker News: GitHub to Disable npm Install Scripts by Default to Stop Supply Chain Attacks, 2026-06-11. https://thehackernews.com/2026/06/github-to-disable-npm-install-scripts.html
- [4] npm documentation: `npm approve-scripts`. https://docs.npmjs.com/cli/commands/npm-approve-scripts
- [5] npm documentation: `npm deny-scripts`. https://docs.npmjs.com/cli/commands/npm-deny-scripts
- [6] npm documentation: `allow-scripts` configuration. https://docs.npmjs.com/cli/using-npm/config#allow-scripts